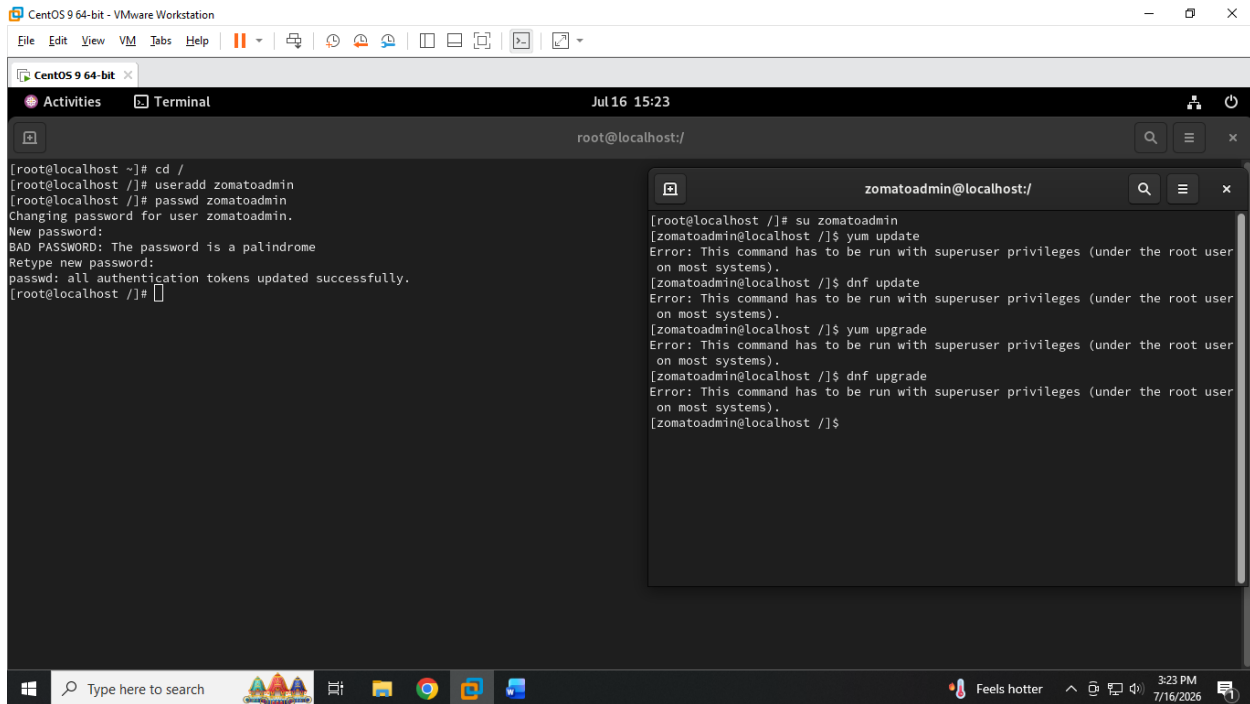


## MODULE-9 : ENHANCED USER SECURITY WITH SUDO

GUIDED BY : VIPUL MIRTRI

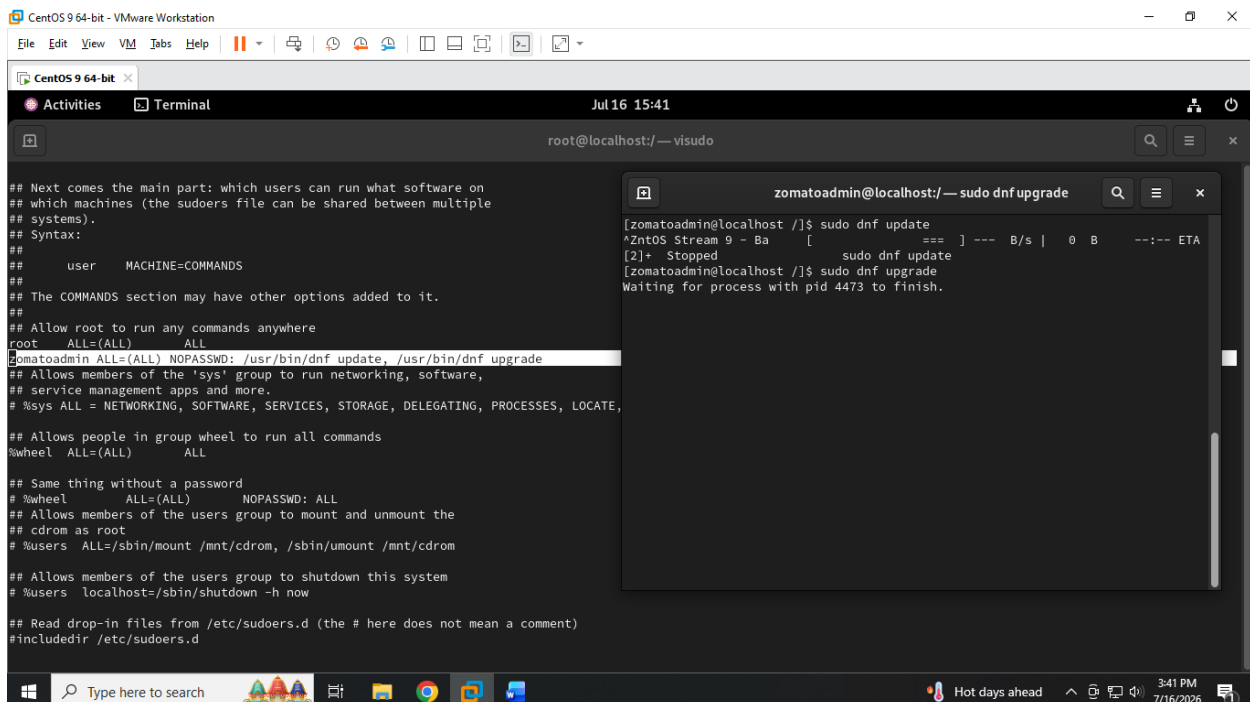
Q.1 Create a new user account called zomatoadmin on your Linux system, and configure it so that this user can run only the apt-get update and apt-get upgrade commands using sudo without entering a password. Hint: Edit the /etc/sudoers file using visudo and use the NOPASSWD directive for specific commands.

ANS :



```
[root@localhost ~]# cd /
[root@localhost /]# useradd zomatoadmin
[root@localhost /]# passwd zomatoadmin
Changing password for user zomatoadmin.
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost /]#
```

```
[root@localhost /]# su zomatoadmin
[zomatoadmin@localhost /]$ yum update
Error: This command has to be run with superuser privileges (under the root user on most systems).
[zomatoadmin@localhost /]$ dnf update
Error: This command has to be run with superuser privileges (under the root user on most systems).
[zomatoadmin@localhost /]$ yum upgrade
Error: This command has to be run with superuser privileges (under the root user on most systems).
[zomatoadmin@localhost /]$ dnf upgrade
Error: This command has to be run with superuser privileges (under the root user on most systems).
[zomatoadmin@localhost /]$
```



```
## Next comes the main part: which users can run what software on
## which machines (the sudoers file can be shared between multiple
## systems).
## Syntax:
##      user    MACHINE=COMMANDS
##
## The COMMANDS section may have other options added to it.
##
## Allow root to run any commands anywhere
root    ALL=(ALL)    ALL
zomatoadmin ALL=(ALL) NOPASSWD: /usr/bin/dnf update, /usr/bin/dnf upgrade
## Allows members of the 'sys' group to run networking, software,
## service management apps and more.
%sys    ALL = NETWORKING, SOFTWARE, SERVICES, STORAGE, DELEGATING, PROCESSES, LOCATE,
## Allows people in group wheel to run all commands
%wheel  ALL=(ALL)    ALL
## Same thing without a password
%wheel  ALL=(ALL)    NOPASSWD: ALL
## Allows members of the users group to mount and unmount the
## cdrom as root
%users  ALL=/sbin/mount /mnt/cdrom, /sbin/umount /mnt/cdrom
## Allows members of the users group to shutdown this system
%users  localhost=/sbin/shutdown -h now

## Read drop-in files from /etc/sudoers.d (the # here does not mean a comment)
#include_dir /etc/sudoers.d
```

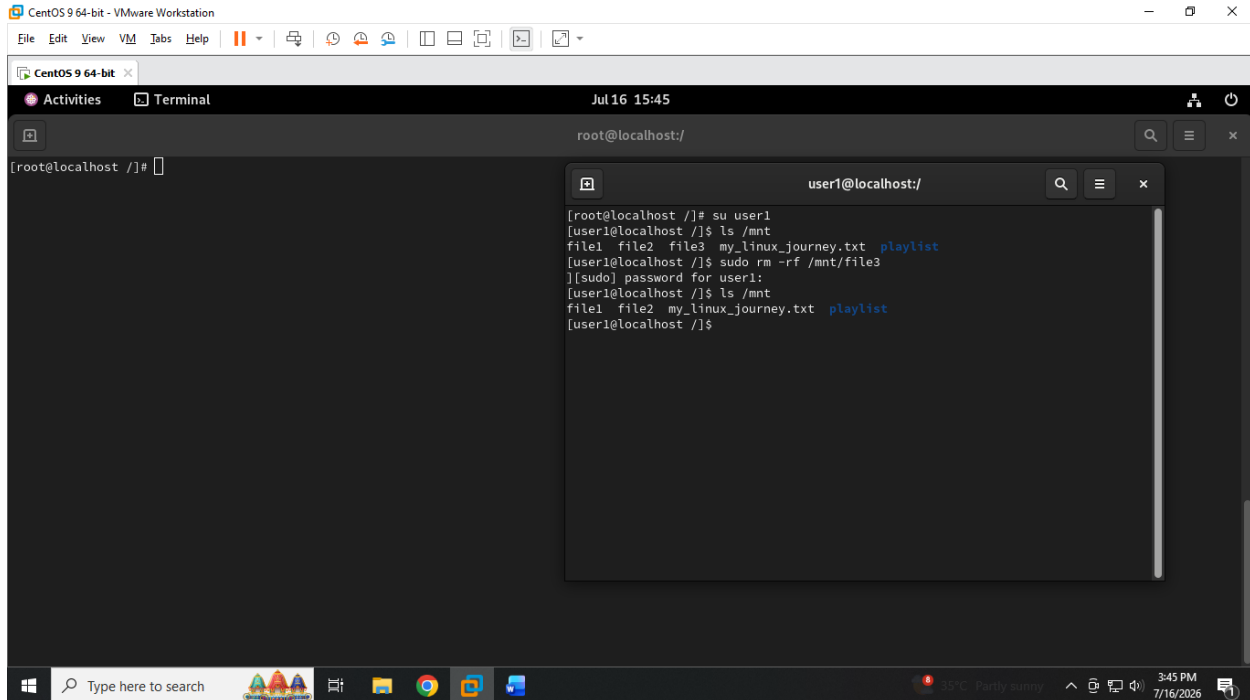
```
[zomatoadmin@localhost /]$ sudo dnf upgrade
^Z
[2]+  Stopped                  sudo dnf update
[zomatoadmin@localhost /]$ sudo dnf upgrade
Waiting for process with pid 4473 to finish.
```

## MODULE-9 : ENHANCED USER SECURITY WITH SUDO

GUIDED BY : VIPUL MIRTRI

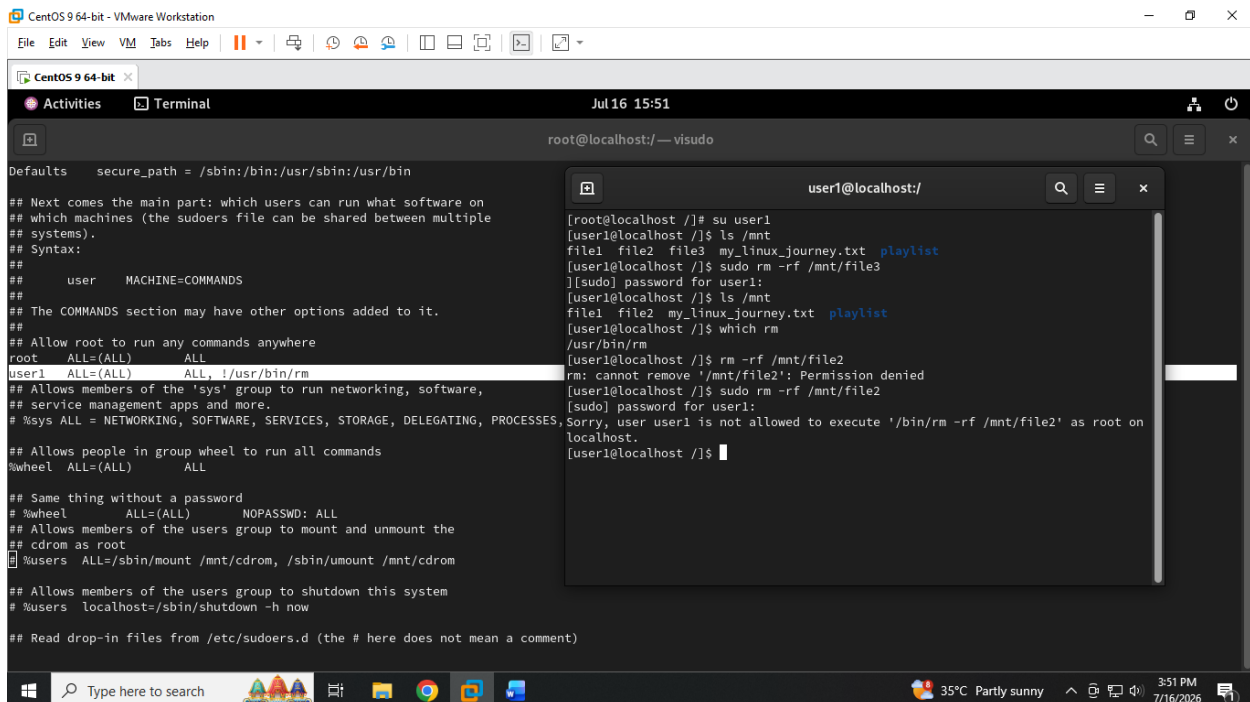
Q.2 Modify the sudo privileges for your existing user so that they cannot run `sudo rm -rf /` but can run all other sudo commands. Constraint: Use command-specific restrictions in the sudoers file to block only the dangerous command.

ANS :



The screenshot shows a terminal window titled "CentOS 9 64-bit - VMware Workstation". The main terminal is running as root at localhost. A secondary terminal window is open, showing the process of switching to user1 and testing sudo privileges. The commands executed are:

```
[root@localhost /]# su user1
[user1@localhost /]$ ls /mnt
file1 file2 file3 my_linux_journey.txt playlist
[user1@localhost /]$ sudo rm -rf /mnt/file3
[sudo] password for user1:
[user1@localhost /]$ ls /mnt
file1 file2 my_linux_journey.txt playlist
[user1@localhost /]$
```



The screenshot shows a terminal window titled "CentOS 9 64-bit - VMware Workstation". The main terminal is running as root at localhost, editing the sudoers file with visudo. The secondary terminal window shows the process of switching to user1 and testing the updated sudo privileges. The commands executed are:

```
[root@localhost /]# su user1
[user1@localhost /]$ ls /mnt
file1 file2 file3 my_linux_journey.txt playlist
[user1@localhost /]$ sudo rm -rf /mnt/file3
[sudo] password for user1:
[user1@localhost /]$ ls /mnt
file1 file2 my_linux_journey.txt playlist
[user1@localhost /]$ which rm
/usr/bin/rm
[user1@localhost /]$ rm -rf /mnt/file2
rm: cannot remove '/mnt/file2': Permission denied
[user1@localhost /]$ sudo rm -rf /mnt/file2
[sudo] password for user1:
Sorry, user user1 is not allowed to execute '/bin/rm -rf /mnt/file2' as root on localhost.
[user1@localhost /]$
```

The sudoers file is being edited with visudo. The relevant lines are:

```
Defaults    secure_path = /sbin:/bin:/usr/sbin:/usr/bin

## Next comes the main part: which users can run what software on
## which machines (the sudoers file can be shared between multiple
## systems).
## Syntax:
##
##      user    MACHINE=COMMANDS
##
## The COMMANDS section may have other options added to it.
##
## Allow root to run any commands anywhere
root    ALL=(ALL)    ALL

user1    ALL=(ALL)    ALL, !usr/bin/rm

## Allows members of the 'sys' group to run networking, software,
## service management apps and more.
## %sys ALL = NETWORKING, SOFTWARE, SERVICES, STORAGE, DELEGATING, PROCESSES,
##
## Allows people in group wheel to run all commands
%wheel  ALL=(ALL)    ALL

## Same thing without a password
## %wheel    ALL=(ALL)    NOPASSWD: ALL
## Allows members of the users group to mount and unmount the
## cdrom as root
## %users    ALL=/sbin/mount /mnt/cdrom, /sbin/umount /mnt/cdrom
##
## Allows members of the users group to shutdown this system
## %users    localhost=/sbin/shutdown -h now

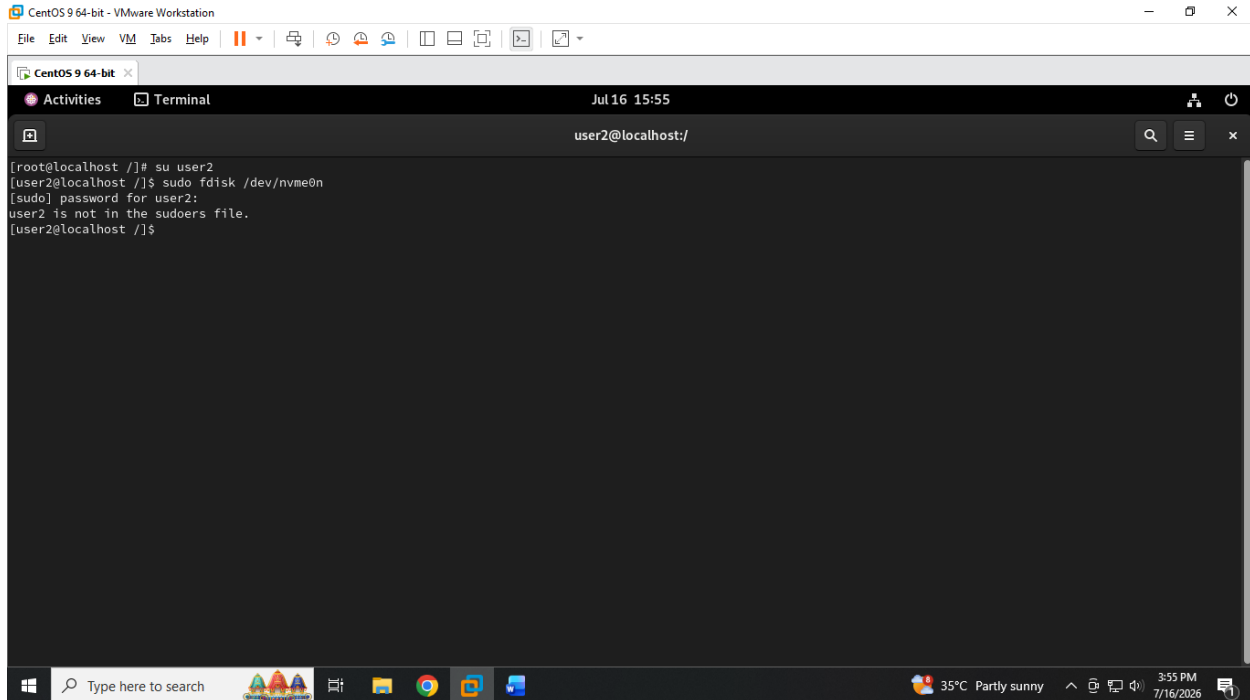
## Read drop-in files from /etc/sudoers.d (the # here does not mean a comment)
```

## MODULE-9 : ENHANCED USER SECURITY WITH SUDO

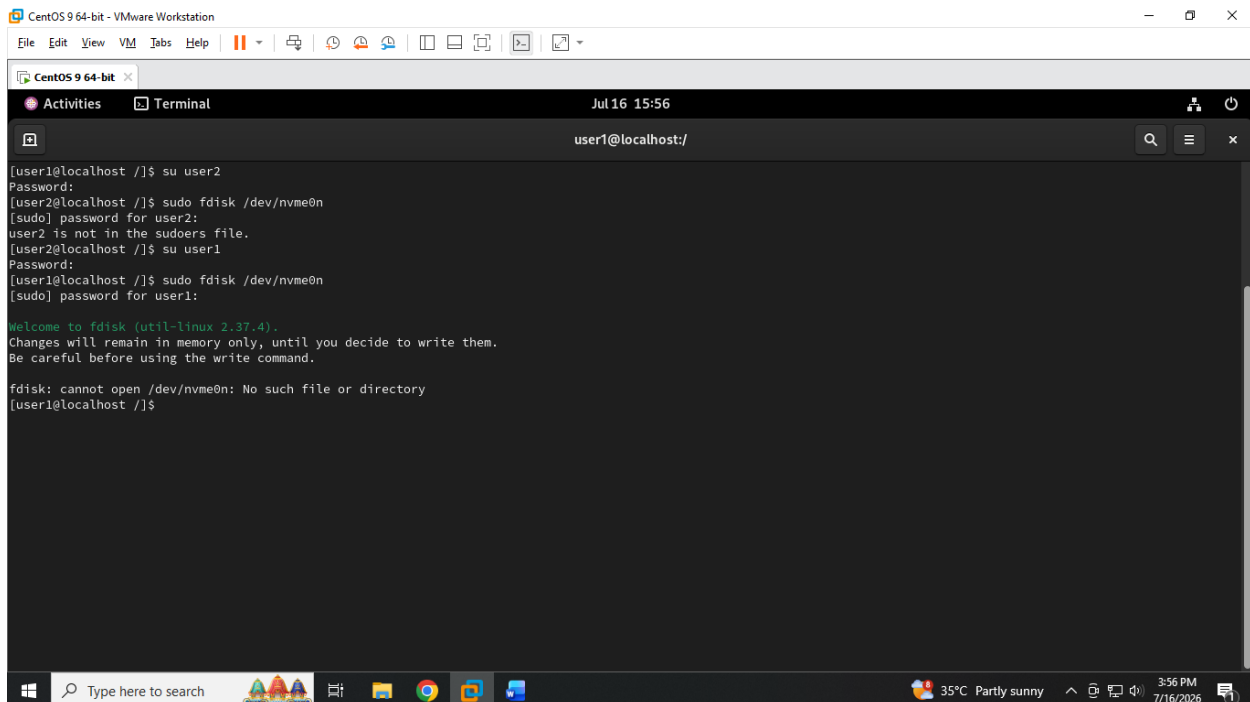
GUIDED BY : VIPUL MIRTRI

Q.3 Simulate a scenario where a user tries to run a command with sudo that they are not allowed to. Take a screenshot of the error message and explain in 2-3 lines why the error occurs based on your sudoers configuration.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 16 15:55
user2@localhost:/
[root@localhost ~]# su user2
[user2@localhost ~]$ sudo fdisk /dev/nvme0n
[sudo] password for user2:
user2 is not in the sudoers file.
[user2@localhost ~]$
```

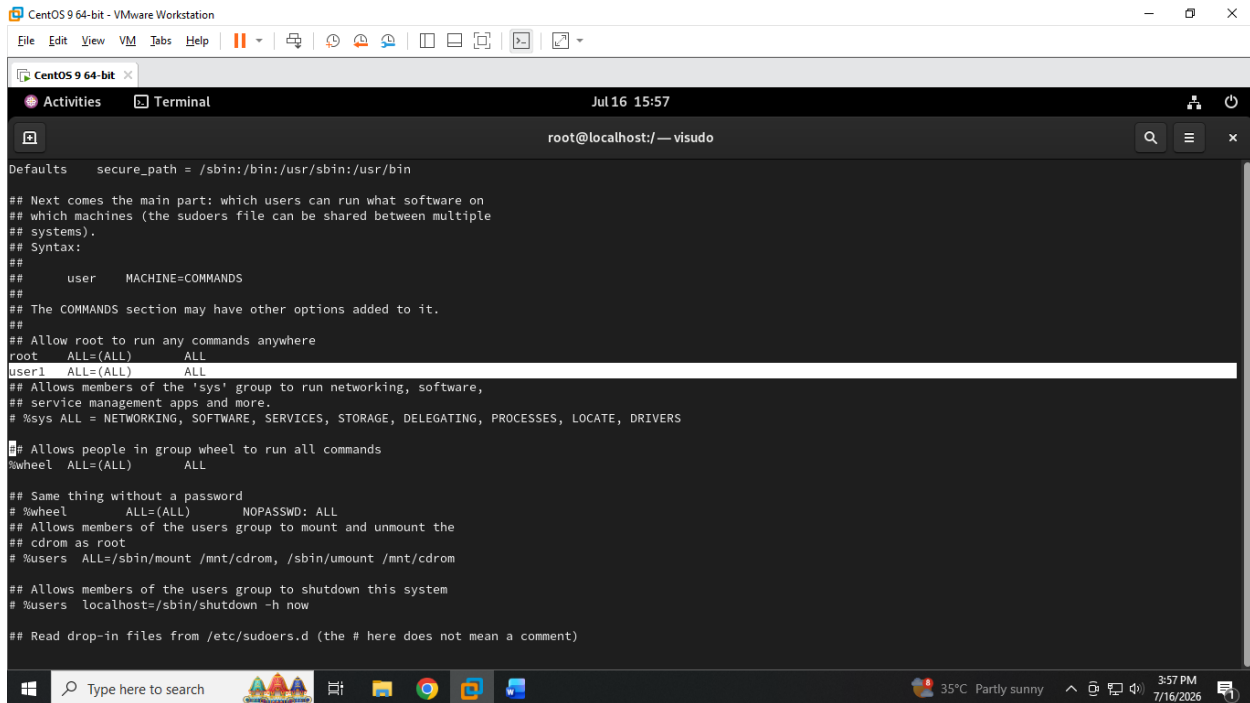


```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 16 15:56
user1@localhost:/
[user1@localhost ~]$ su user2
Password:
[user2@localhost ~]$ sudo fdisk /dev/nvme0n
[sudo] password for user2:
user2 is not in the sudoers file.
[user2@localhost ~]$ su user1
Password:
[user1@localhost ~]$ sudo fdisk /dev/nvme0n
[sudo] password for user1:
Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

fdisk: cannot open /dev/nvme0n: No such file or directory
[user1@localhost ~]$
```

## MODULE-9 : ENHANCED USER SECURITY WITH SUDO

GUIDED BY : VIPUL MIRTRI



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 16 15:57
root@localhost:~# visudo

Defaults    secure_path = /sbin:/bin:/usr/sbin:/usr/bin

## Next comes the main part: which users can run what software on
## which machines (the sudoers file can be shared between multiple
## systems).
## Syntax:
##
##    user    MACHINE=COMMANDS
##
## The COMMANDS section may have other options added to it.
##
## Allow root to run any commands anywhere
root    ALL=(ALL)    ALL
user1   ALL=(ALL)    ALL

## Allows members of the 'sys' group to run networking, software,
## service management apps and more.
# %sys ALL = NETWORKING, SOFTWARE, SERVICES, STORAGE, DELEGATING, PROCESSES, LOCATE, DRIVERS

## Allows people in group wheel to run all commands
%wheel  ALL=(ALL)    ALL

## Same thing without a password
# %wheel    ALL=(ALL)    NOPASSWD: ALL
## Allows members of the users group to mount and unmount the
## cdrom as root
# %users    ALL=/sbin/mount /mnt/cdrom, /sbin/umount /mnt/cdrom

## Allows members of the users group to shutdown this system
# %users    localhost=/sbin/shutdown -h now

## Read drop-in files from /etc/sudoers.d (the # here does not mean a comment)
```

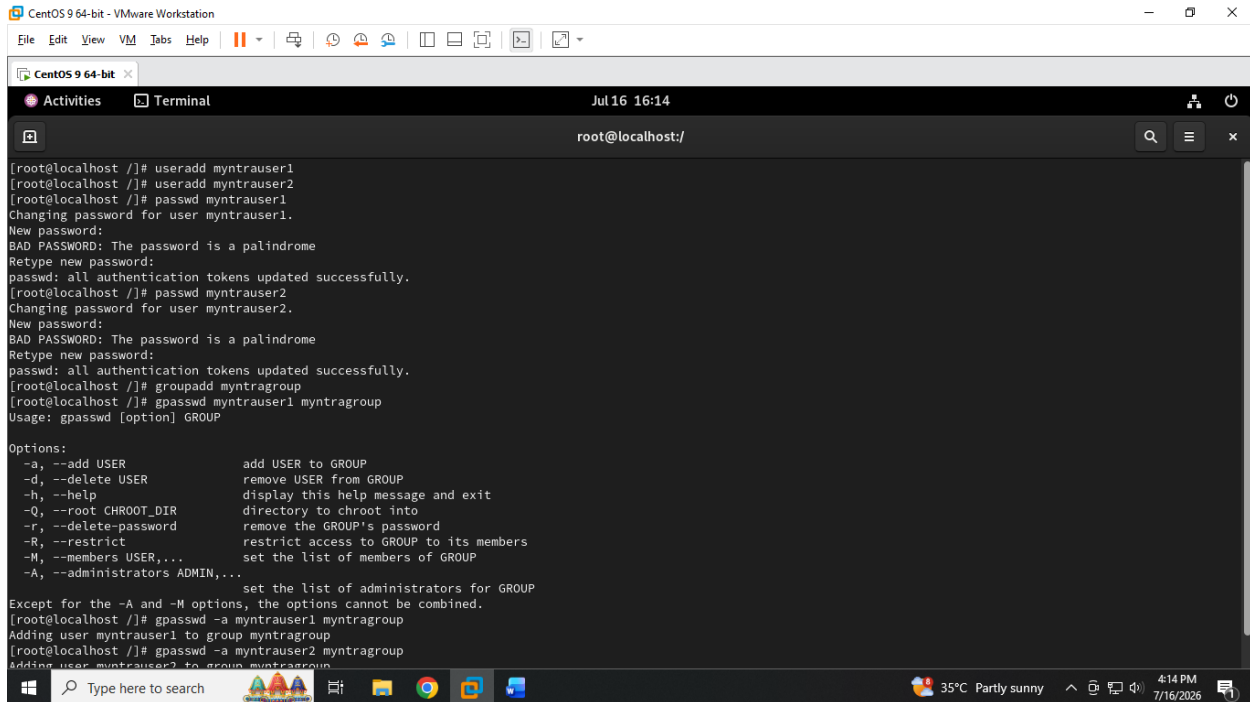
EXPLANATION : as you can see in img.1: user2 is trying to use fdisk command with sudo privilege but it can not use command. And when user1 do the same it can use command normally, its all because user2 is not in the “sudoers.” As you can see img.3: only user1 have the authority to use all sudo privilege commands.

## MODULE-9 : ENHANCED USER SECURITY WITH SUDO

GUIDED BY : VIPUL MIRTRI

Q.4 Set up a group called myntragroup and add two users to it. Grant sudo privileges to this group so that all members can restart the networking service (systemctl restart networking) but cannot edit any files in /etc.

ANS :

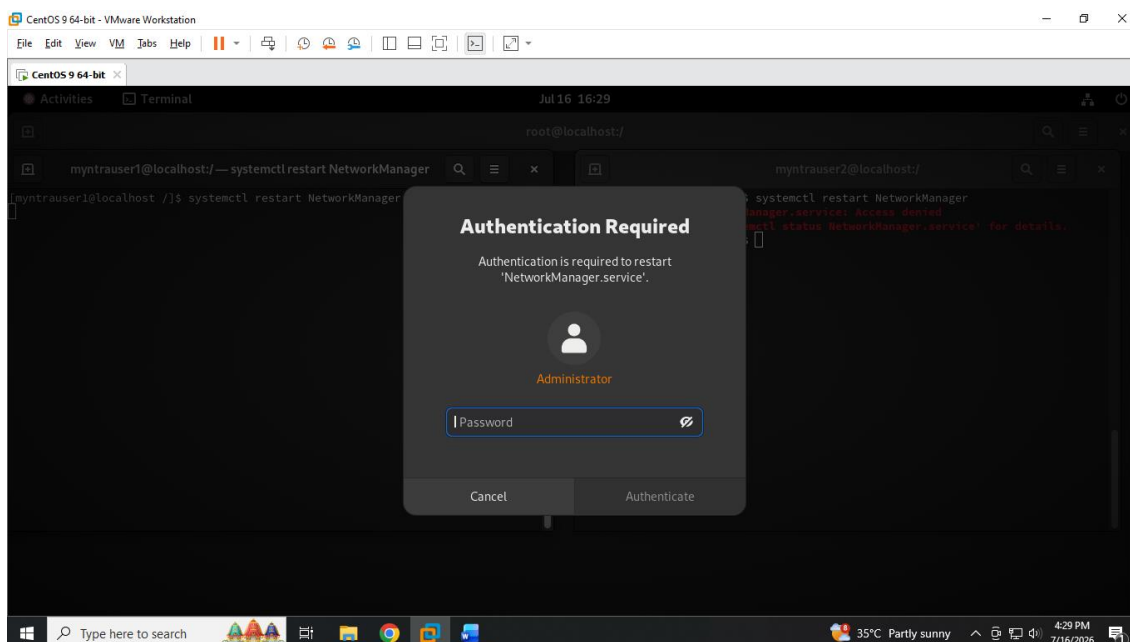


```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 16 16:14
root@localhost:/

[root@localhost ~]# useradd myntrauser1
[root@localhost ~]# useradd myntrauser2
[root@localhost ~]# passwd myntrauser1
Changing password for user myntrauser1.
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# passwd myntrauser2
Changing password for user myntrauser2.
New password:
BAD PASSWORD: The password is a palindrome
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost ~]# groupadd myntragroup
[root@localhost ~]# gpasswd myntrauser1 myntragroup
Usage: gpasswd [option] GROUP

Options:
-a, --add USER          add USER to GROUP
-d, --delete USER       remove USER from GROUP
-h, --help              display this help message and exit
-Q, --root CHROOT_DIR   directory to chroot into
-r, --delete-password    remove the GROUP's password
-R, --restrict          restrict access to GROUP to its members
-M, --members USER,... set the list of members of GROUP
-A, --administrators ADMIN,... set the list of administrators for GROUP

Except for the -A and -M options, the options cannot be combined.
[root@localhost ~]# gpasswd -a myntrauser1 myntragroup
Adding user myntrauser1 to group myntragroup
[root@localhost ~]# gpasswd -a myntrauser2 myntragroup
Adding user myntrauser2 to group myntragroup
```



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 16 16:29
root@localhost:/

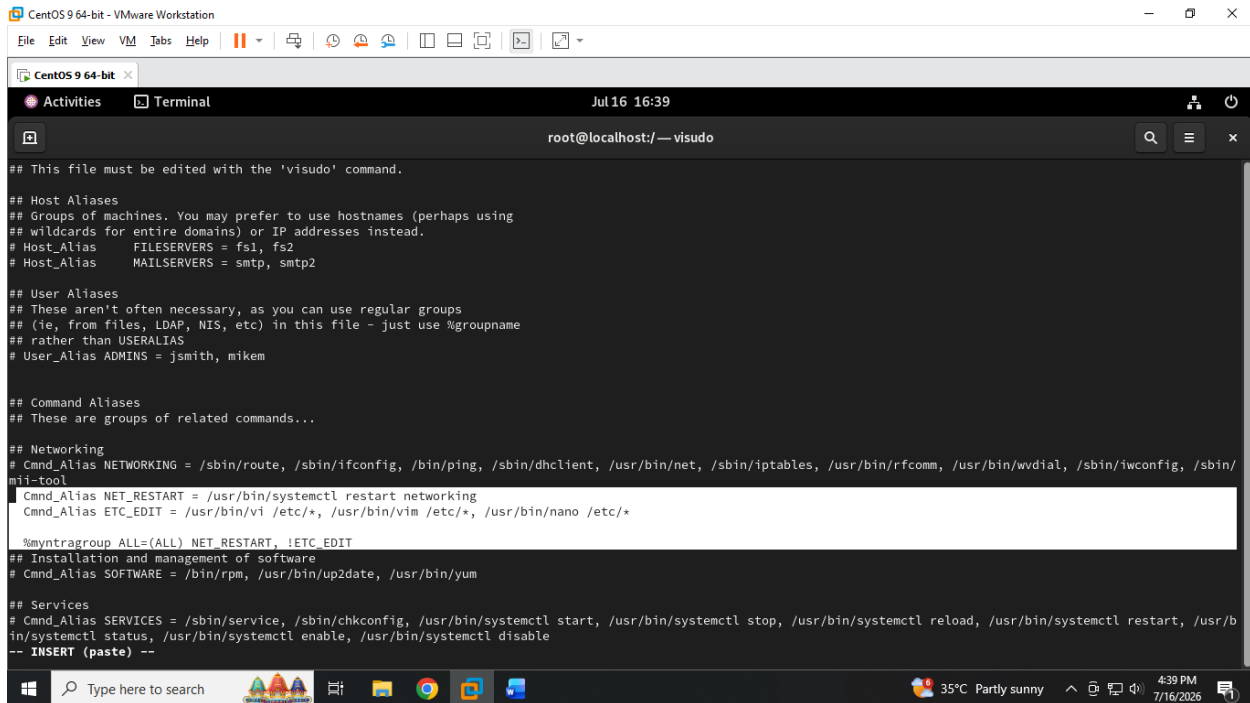
myntrauser1@localhost:~$ systemctl restart NetworkManager
Authentication Required
Authentication is required to restart 'NetworkManager.service'.
Administrator
Password
Cancel Authenticate

myntrauser2@localhost:~$ systemctl restart NetworkManager
NetworkManager.service: Access Denied
journal status 'NetworkManager.service' for details.
```

As you can see when I enter command to restart NetworkManager it will ask me for administrator password , because till now I have not give them authorization to run this command using sudo privilege .

## MODULE-9 : ENHANCED USER SECURITY WITH SUDO

GUIDED BY : VIPUL MIRTRI



The screenshot shows a terminal window titled 'CentOS 9 64-bit - VMware Workstation'. The terminal is running the 'visudo' command, editing the /etc/sudoers file. The file content is as follows:

```
## This file must be edited with the 'visudo' command.

## Host Aliases
## Groups of machines. You may prefer to use hostnames (perhaps using
## wildcards for entire domains) or IP addresses instead.
# Host_Alias   FILESERVERS = fs1, fs2
# Host_Alias   MAILSERVERS = smtp, smtp2

## User Aliases
## These aren't often necessary, as you can use regular groups
## (ie, from files, LDAP, NIS, etc) in this file - just use %groupname
## rather than USERALIAS
# User_Alias ADMINS = jsmith, mikem

## Command Aliases
## These are groups of related commands...

## Networking
# Cmnd_Alias NETWORKING = /sbin/route, /sbin/ifconfig, /bin/ping, /sbin/dhclient, /usr/bin/net, /sbin/iptables, /usr/bin/rfcomm, /usr/bin/wvdial, /sbin/iwconfig, /sbin/mii-tool
# Cmnd_Alias NET_RESTART = /usr/bin/systemctl restart networking
# Cmnd_Alias ETC_EDIT = /usr/bin/vi /etc/*, /usr/bin/vim /etc/*, /usr/bin/nano /etc/*

%myntgroup ALL=(ALL) NET_RESTART, !ETC_EDIT

## Installation and management of software
# Cmnd_Alias SOFTWARE = /bin/rpm, /usr/bin/up2date, /usr/bin/yum

## Services
# Cmnd_Alias SERVICES = /sbin/service, /sbin/chkconfig, /usr/bin/systemctl start, /usr/bin/systemctl stop, /usr/bin/systemctl reload, /usr/bin/systemctl restart, /usr/bin/systemctl status, /usr/bin/systemctl enable, /usr/bin/systemctl disable
-- INSERT (paste) --
```

The terminal window also shows the system clock as 'Jul 16 16:39' and the user as 'root@localhost:~'. The bottom of the window shows a Windows taskbar with a search bar and various application icons.

After entry in visudo the both the users and now use this command. And can not edit /etc.

## MODULE-9 : ENHANCED USER SECURITY WITH SUDO

GUIDED BY : VIPUL MIRTTRI

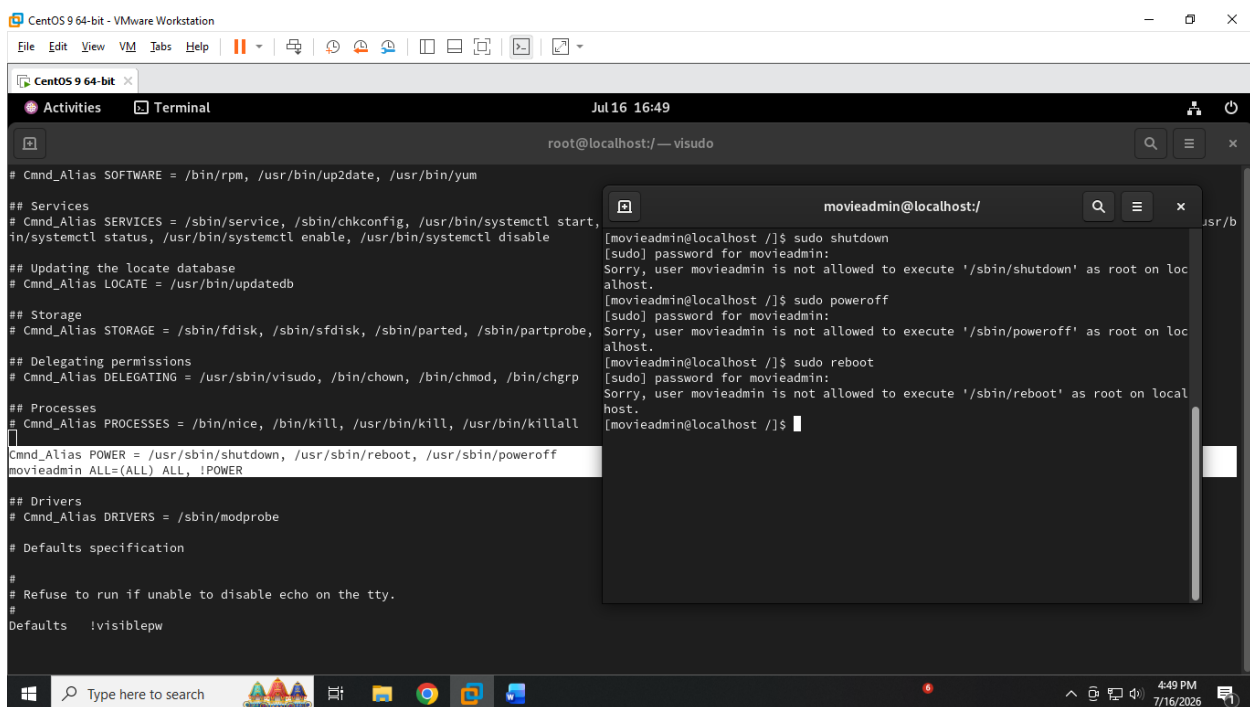
Q.5 Use ChatGPT to generate a sample /etc/sudoers entry that allows a user called movieadmin to run all commands as root except for shutting down or rebooting the system. Copy the AI's suggestion, test it on your system, and write one line about whether it worked as expected.

ANS :

A suitable /etc/sudoers entry is:

Cmd\_Alias POWER = /usr/sbin/shutdown, /usr/sbin/reboot, /usr/bin/systemctl poweroff, /usr/bin/systemctl reboot

movieadmin ALL=(ALL) ALL, !POWER



The screenshot shows a terminal window titled 'CentOS 9 64-bit - VMware Workstation' with a terminal window open. The terminal shows the contents of the /etc/sudoers file, which includes various aliases and permissions. The relevant entry for movieadmin is highlighted: `movieadmin ALL=(ALL) ALL, !POWER`. To the right, a separate terminal window shows the movieadmin user attempting to run `sudo shutdown`, `sudo poweroff`, and `sudo reboot`, all of which are denied with the message: 'Sorry, user movieadmin is not allowed to execute ... as root on localhost.'

```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Jul 16 16:49
root@localhost:~# visudo
# Cmnd_Alias SOFTWARE = /bin/rpm, /usr/bin/up2date, /usr/bin/yum
## Services
# Cmnd_Alias SERVICES = /sbin/service, /sbin/chkconfig, /usr/bin/systemctl start,
in/systemctl status, /usr/bin/systemctl enable, /usr/bin/systemctl disable
## Updating the locate database
# Cmnd_Alias LOCATE = /usr/bin/updatedb
## Storage
# Cmnd_Alias STORAGE = /sbin/fdisk, /sbin/sfdisk, /sbin/parted, /sbin/partprobe,
## Delegating permissions
# Cmnd_Alias DELEGATING = /usr/sbin/visudo, /bin/chown, /bin/chmod, /bin/chgrp
## Processes
# Cmnd_Alias PROCESSES = /bin/nice, /bin/kill, /usr/bin/kill, /usr/bin/killall
Cmnd_Alias POWER = /usr/sbin/shutdown, /usr/sbin/reboot, /usr/sbin/poweroff
movieadmin ALL=(ALL) ALL, !POWER
## Drivers
# Cmnd_Alias DRIVERS = /sbin/modprobe
# Defaults specification
#
# Refuse to run if unable to disable echo on the tty.
#
Defaults !visiblepw
movieadmin@localhost:/]$ sudo shutdown
[sudo] password for movieadmin:
Sorry, user movieadmin is not allowed to execute '/sbin/shutdown' as root on loc
alhost.
[movieadmin@localhost /]$ sudo poweroff
[sudo] password for movieadmin:
Sorry, user movieadmin is not allowed to execute '/sbin/poweroff' as root on loc
alhost.
[movieadmin@localhost /]$ sudo reboot
[sudo] password for movieadmin:
Sorry, user movieadmin is not allowed to execute '/sbin/reboot' as root on local
host.
[movieadmin@localhost /]$
```